Documentation for Microvilli Analysis Script

Script processes multi-layer TIFF image stacks, detect microvilli, extract intensity profiles, and calculate background intensities. Results are visualized and saved in CSV format for further analysis.

Preprocessing data with Fiji

Before running the script, you need to prepare the images using Fiji (ImageJ).

Script only runs for single cell images. Need to be cut at the end at some point.

1. **Merge Channels into a single TIFF file**:
    - open Fiji and merge your two color channels into one TIFF stack
        - **first channel** should contain the Actin or membrane stain that outlines the microvilli
        - **second channel** should contain the antibody stain
2. **You can perform a background subtraction with fiji, but don't need to because the script does one too**
3. **Determine the Start and Center Layers**:
    - identify the **surface layer** (where the substrate or surface begins) in the stack.

      This is your `start_layer`

      (theres a trick for this but **it doesnt work all the time,**

      the second channel/ antibody channel usually has a peak in background when youre at the substrate surface layer, check using "Images" -> "Stacks" -> "Plot z-Axis Profile", choose the second channel and select a background area near your cell (before cutting the image …))
    - identify the **z-layer** that shows the microvilli most clearly. This is your `center_z`
    - avoid using the same layer as the start_layer for this. It often contains the membrane ring, which can interfere with microvilli detection in later steps
4. **Shift Correction not needed bc the script does one too**

Using the Script

1. Setting Up the Script

before running the script install necessary Python dependencies

```
pip install imageio
```

```
pip install numpy
```

```
pip install matplotlib
```

```
pip install scipy
```

```
pip install scikit-image tkinter
```

2. Configure the Script

There's a `CONFIG` dictionary. Configure the parameters for your specific data

- **`center_z`**: z-layer where microvilli are most visible (from Fiji analysis).
- **`start_layer`**: z-layer where the substrate or surface begins (from Fiji analysis).
- **`sigma`**: radius used for blob (microvilli) detection.
- **`intensity_threshold`**: minimum intensity for a blob to be considered valid.
- **`percentile_threshold`**: threshold used to define the cell mask.
- **`z_spacing`**: distance between z-layers in micrometers (µm).
- **`rolling_ball_radius`**: radius for background subtraction (rolling ball filter).
- **`tolerance_factor`**: parameter helps filter intensity profiles based on background levels. preferably stays the same for one experiment (should def stay the same for one condition).
- **`shift_correction_enabled = False`**: (or true) depends on if you need one or not.

Set these values based on your data and experiment. the underlined ones have to be set before running. Rest can stay on default for the first run.

3. Running the Script

To run the script:

1. open spyder and load in the latest script “Asya_detection_script”

2. **before running the script,** enter your “`center_z`” and “`start_layer`” values

3. if you need a shift correction, change the line “`shift_correction_enabled = False`” to “`shift_correction_enabled = True`”

4. **run the script**

5. **open the TIFF file**:
   when you run the script, a file dialog will appear allowing you to select the TIFF stack you created in Fiji.

6. **two images will be created.** If the `intensity_threshold` is okay, the second image should now show your cell (channel1) at center_z, with spots marking the detected microvilli (MV), referred to as blobs**.** The red circle (`sigma`) should nicely/ tightly fit the size of your MV.. this is important because the lengths are depending on that.

   - **if you get an error, you need to set your `intensity_threshold` LOWER**
   - if you see way too many red circles, you need to raise it
   - run again, until you’re satisfied with the selection, preferably shouldn’t vary too much between cells in the same condition.

   the first image

   - the first shows the detected blobs in the image
   - the second shows the image after the rolling ball filter
   - the third shows the cell mask. Blue area should be your cell. the gray area should be your blobs and the substrate (corners). this you can modify with `percentile_threshold` (1-100, but 80-90 is usually fine for all data)

   the second image

   - top left shows the intensity profiles of the blobs across the z-stack
   - top right shows an histogram of the lengths
   - bottom left is the selected MV’s
   - bottom right bar chart of the lengths
   - code also prints out the actual length values

7. **what the script did:**

   - MV are detected in the `center_z` **l**ayer using the parameters set in the configuration (`sigma` and `intensity_threshold`)

- once the blobs were detected, the script extracted intensity profiles across all z-layers. These profiles represent how the intensity of the detected microvilli changes across the stack

- the script filters out all profiles that are just 1 frame long, since the mess up the average length calculation a lot

8. **Saving Results**:
the extracted intensity profiles and metadata (such as z-spacing) are saved as CSV files in the file folder for further analysis. The CSV files can be analyzed with another script.

9. Further Analysis:

the actual analysis is done with the other script “Asya_CSV_analysis”

for that, copy the path of the condition folders and paste them into the first few lines in code

ERRORs:

“runfile('C:/Users/bio_polatasy/Desktop/XXXXXX_FINAL_CODE_ROLLING_BALL.py', wdir='C:/Users/bio_polatasy/Desktop')

TIFF stack shape: (23, 2, 124, 116)

Blobs detected: 0

No blobs detected. Try adjusting the parameters.

Traceback (most recent call last):

File ~\Desktop\XXXXXX_FINAL_CODE_ROLLING_BALL.py:265 in <module>

exit()

NameError: name 'exit' is not define”

—> intensity_treshhold is too high.

“runfile('C:/Users/bio_polatasy/Desktop/XXXXXX_FINAL_CODE_ROLLING_BALL.py', wdir='C:/Users/bio_polatasy/Desktop')

TIFF stack shape: (23, 2, 124, 116)

Invalid z-layer index. Exiting.

Traceback (most recent call last):

File ~\Desktop\XXXXXX_FINAL_CODE_ROLLING_BALL.py:229 in <module>

exit()

NameError: name 'exit' is not defined”

—> center or start_layer input is wrong (out of range)